

Strategy White Paper

Image-Based Sexual Abuse

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1 Image-Based Sexual Abuse

Image-based sexual abuse, as defined by the Australian Institute of Criminology (2022, p. 1), refers to the “non-consensual creation, distribution or threatened distribution of nude or sexual images”. According to the institute, revenge pornography is the main form of IBSA reported on by the media, and is primarily driven by a desire for revenge or control. The broader category of image-based sexual abuse is said to be motivated by factors such as voyeurism, sexual gratification, or the pursuit of monetary or reputational gain. Cases driven by revenge or control typically involve intimate partners, friends, family members or others in close social proximity to the abuser. However, IBSA can also involve offenders who are unknown to the victims (AIC 2022).

Victimisation rates for image-based sexual abuse in Australia differ according to various studies. Although both surveys referenced in the studies were conducted between 2016–2017, RMIT’s (2020) study indicates a victimisation rate of 35.2%, whereas AIC’s (2022) report suggests a lower rate of 23%. Both studies however, agree that these figures likely underestimate the prevalence, stating that a high proportion of victims do not discover that sexual imagery of them has been created or distributed. Furthermore, when considering gender, both sources highlight that women are disproportionately affected, with significantly higher victimisation rates than men. Moreover, AIC identifies vulnerable groups, including Indigenous people, LGBTQI+ individuals, those with physical or mental disabilities, and minors, as particularly at risk. Among these, people with disabilities reported the highest risk of IBSA, with 56% of respondents disclosing a disability reporting to have experienced some form of image-based sexual abuse.

Regarding perpetrators, RMIT (2020) states that 10% of Australians have engaged in IBSA. Additionally, according to AIC’s survey (2022, p. 13), 54% of offenders were male, 33% were female, and the remaining 13% have an unknown gender. AIC (2022) further notes that IBSA is predominantly carried out through messaging apps and social media platforms, with a smaller yet significant number of incidents involving websites dedicated to revenge pornography. While some individuals may consensually send nude or sexual images of themselves to others through these platforms, the ease of downloading and sharing these images often leads to their non-consensual distribution. The rise of Artificial Intelligence in recent years has also drastically simplified the perpetration of IBSA, with deepfake technology enabling users to create sexual imagery by placing the victims’ faces onto other bodies (Okolie 2023). This technology, increasingly accessible through emerging mobile apps and websites offering deepfake creation for a monthly subscription, may become a major

pathway to offending (DHS 2022).

Victims of image based sexual abuse experience emotional and psychological trauma, including anxiety, depression, and social isolation (Okolie 2023). Additionally, they may face societal victim-shaming, disruptions to education or their careers, and, in severe cases, post-traumatic stress disorder and an increased risk of suicide (Klein & Prabhakar 2024). This underscores the need for robust preventative measures that can combat IBSA in the constantly evolving cyberspace.

2 Situational Crime Prevention

Among the various strategies for crime prevention, situational crime prevention (SCP) is an approach based on the rational choice and routine activity theories, where settings and circumstances that facilitate criminal activities are prioritised, instead of focusing solely on the offenders themselves (Clarke 1983; Clarke 2009; Ho et al. 2022). Considering the notion that offenders make calculated decisions to commit crimes based on potential risks and rewards, it asserts that by “increasing effort, elevating the risks, reducing rewards, minimizing provocations and removing excuses”, the likelihood of crime can be significantly reduced, making the targeted activity less appealing and more difficult to carry out (Cornish & Clarke 2003; Ho et al. 2022).

While SCP was originally created to reduce and prevent the occurrence of offline crimes, it has also proven to be useful in combating cybercrime (Russel et al. 2019). One of its strengths is the immediate impact it can have on crime prevention. For example, if an organisation were to increase the effort required for security breaches by adding multi-factor authentication to its network, it will immediately reduce cyber-attacks related to unauthorised access (Weidman & Grossklags 2017). Similarly, deploying antivirus programs and firewalls will immediately reduce the risk of malware infecting their network (Algaith et al. 2016). Furthermore, these same measures can be applied across various digital environments such as personal devices and critical infrastructure, proving the broad applicability of SCP. Using a SCP approach to cybercrime prevention can ultimately reduce the fear of cyber-attacks among users and organisations, which is essential for protecting critical infrastructure, as well as promoting user trust in online systems and applications (Russel et al. 2019).

However, SCP also has limitations in cybercrime prevention. One major concern is the ethical implications of increased surveillance. Data gathering and monitoring of online activities to enhance crime detection (which is under the pillar of ‘increasing the risk’ in SCP) raises ethical issues about privacy and civil rights (Eivazi

2011). Therefore, certain measures under the SCP framework may create a digital environment where users do not trust organisations with data privacy and resort to technology that increases anonymity, which would in turn make crime detection more difficult (Clarke 1995).

Furthermore, implementing and maintaining effective SCP measures in cybersecurity can be expensive and even unaffordable for many organisations and individuals (Akansha & Prashant 2017). While a certain software update may fix a certain vulnerability in a system, cybercriminals constantly adapt and find new ways to exploit other vulnerabilities, which prompts the need for constant, ongoing security improvements and adaptation. Furthermore, as affirmed by the above authors, it may not be financially viable to spend resources to sustain the security of an item or asset if that expense exceeds the value of the item itself. Because of this, smaller businesses and individuals with limited resources may not be able to acquire and sustain new security tools to maintain protection.

Considering both the advantages and limitations, the following section will go over each of the situational crime prevention strategies currently used in preventing image based sexual abuse in the cyberspace.

3 Existing Prevention Strategies

3.1 Increasing the Effort to Offend

In the realm of cybersecurity, the primary measures taken to increase the effort to commit cybercrimes are technological safeguards such as encryption, multi-factor authentication and robust security settings (Russel et al. 2019). These measures can create barriers that make it significantly harder for potential offenders to access or distribute private images.

For instance, social media sites such as Instagram have settings that allows users to prevent others from downloading their images or content (Meta 2024). This can be considered a form of target hardening as it protects individuals from unauthorised distribution of their personal images. In cases where a user with a private profile willingly shares a sexual image of themselves, it could prevent others from downloading and then publicly sharing that image. Certain social media platforms also employ community guidelines that prohibit sharing of sexual imagery. This can both protect victims from inadvertently providing material to potential offenders and prevent offenders from distributing non-consensual images on their platforms (Aguerri, Miró-Llinares & Gómez-Bellví 2023).

Encryption also plays a vital role in protecting data and private images from unauthorised access and distribution. Applications such as Whatsapp provide end-to-end encryption which requires a receiver-specific key to decrypt and read a message, preventing hackers from potentially intercepting communication channels and stealing private images (Meta 2024). However, in cases such as revenge pornography where the perpetrator already has access to the image, encryption may not provide to be useful in preventing IBSA.

Although these measures offer a degree of protection and target-hardening, they may not be able to completely prevent an individual from becoming a victim or a perpetrator of this cybercrime. While prohibiting sharing of sexual imagery under community guidelines can deter some offenders, they may resort to alternative platforms that allow sexual imagery or other secure websites and networks. Moreover, user compliance issues, as well as ethical issues from social media sites monitoring user activity can further complicate the effectiveness of these measures, as explained in the previous section.

3.2 Increasing the Perceived Risk of Offending

In the context of IBSA, increasing the perceived risk of offending can include various stakeholders, including administrators, moderators, and users, in creating an environment where the likelihood of detection and punishment is high (Flynn, Cama & Scott 2022).

In social media applications and online communities, admins and moderators actively contribute to increasing detection by actively monitoring content and enforcing community guidelines. Community moderators are responsible for removing and reporting non-consensual sexual imagery, which creates an environment where sharing or distribution of such content is not tolerated (Flynn, Cama & Scott 2022). Additionally, users can also contribute to increasing perceived risk by using reporting mechanisms to report suspected IBSA to law enforcement or social media organisations, triggering investigations and potential legal action against offenders. However, a significant limitation here is that users other than the victim may find it challenging to determine whether an image was shared consensually, which may lead to underreporting or misreporting of incidents.

An overarching limitation is that this strategy may not be applicable to the dark-web and sites specifically dedicated to revenge pornography and non-consensual material. Such sites operate outside the reach of mainstream regulatory frameworks and do not adhere to the same standards of content moderation, which complicates efforts to increased perceived risk of offending (Choe 2022).

However, a solution to this also exists: honeypot sites designed to attract and monitor potential offenders. They create a sense of resistance or fear when using illegal sites, while enabling law enforcement organisations to investigate and apprehend perpetrators (Kambow & Passi 2014). As discussed earlier, honeypot sites may become particularly useful in increasing the risk of engaging in deepfake pornography creation, especially as this becomes more prevalent with the emergence of web apps utilising generative AI (Klein & Prabhakar 2024). At the same time, honeypot sites must be carefully managed to ensure they do not encourage or facilitate illegal activities.

The ‘Digital footprint’ and user activity tracking, combined with transparency, or informing users that their activity is being logged, can also increase the perceived risk of engaging in criminal activity online (Henry et al. 2020). Individuals may choose to avoid engaging in criminal activity such as IBSA because it can be seen by the administrators or another entity such as their Internet Service Provider. However, since this is a form of digital surveillance, it again raises ethical issues concerning privacy and user monitoring. Furthermore, offenders may use fake identities and anonymizing tools, which can complicate efforts to track activity and potentially remove the risk of being detected.

3.3 Reducing the Reward of Offending

For image-based sexual abuse, especially in revenge pornography cases, reducing the reward for offending can be accomplished by limiting the spread of non-consensual images (Flynn, Cama & Scott 2022). This can involve implementing quick-response mechanisms or algorithms to remove offending content. Governmental organisations currently offers free online tools such as <https://takeitdown.ncmec.org/> and <https://stopncii.org/> that allow victims to remove their images from the internet (eSafety Commissioner 2024). Furthermore, social media applications are able to ban IP addresses of offenders, which prevents them from re-uploading the removed images (Aguerri, Miró-Llinares & Gómez-Bellví 2023).

As for IBSA that is financially motivated, payment processors such as banks and digital payment platforms (PayPal, Wise etc.) are able to monitor transactions for suspicious activity, and as a result, block payments and cut off revenue streams of offenders, making it financially unviable to engage and therefore reduce the motivation to offend (Wolak et al. 2008). Limitations of this approach includes the existence of cryptocurrencies – the anonymity offered by cryptocurrencies such as Bitcoin and Ethereum allow offenders to conduct transactions without the oversight from payment processors or banks (Brown 2016). The level of surveillance required by financial institutions also fuel the rise of these cryptocurrencies, which may in

turn enable offenders.

Another general limitation is that these strategies are reactive other than proactive, addressing the issue after the harm has occurred. While social media platforms and financial institutions can remove or block the offenders, the images may have already been widely shared or saved locally by other users (Abayomi 2020).

3.4 Reducing Offender Provocation

Reducing offender provocation involves creating an environment where potential triggers for offending are minimized (Cornish & Clarke 2003; Ho et al. 2022). These provocations can stem from various sources such as emotional distress from relationships, such as “breakups”, which often lead to retaliation or revenge by sharing private images without consent (AIC 2020).

Compared to the other situational crime prevention techniques, there is currently a limited number of targeted strategies specifically aimed at reducing provocations for IBSA. Online therapy that ensures client confidentiality can provide support for individuals experiencing emotional distress, potentially mitigating the urge to offend. Public awareness campaigns on psychological and legal consequences of IBSA may also help individuals combat provocations (such as peer pressure or subcultural influences) that could lead to offending (Clarke 2009).

Despite these measures, reducing provocation may be particularly challenging in the context of cybercrime. According to Jaishankar (2008, p. 292–296), Individuals who “suppress criminal tendencies in the physical space may be more likely to commit crime in the cyberspace”, and the anonymity and perceived distance provided by the internet allows ex-partners that may appear unprovoked to still offend to relieve “suppressed negative feelings” (Abayomi 2020, p. 632–633). Because of this, it may be difficult to identify potential offenders and effectively intervene before the crime occurs.

3.5 Removing Excuses

Removing excuses involves eliminating justifications that can be used by a potential offender to rationalise their behavior (Cornish & Clarke 2003; Ho et al. 2022). Clear legal definitions have been made publicly available on governmental websites, such as <https://www.esafety.gov.au/>, that remove ambiguity and explicitly criminalize IBSA. Furthermore, most educational institutions host programs that highlight the harm caused by IBSA and promote empathy, which can also play a role in removing excuses (Flynn, Cama & Scott 2022).

One of the main challenges imposed when removing excuses for this cybercrime is that despite clear laws and educational efforts, some individuals may still rationalize their behavior due to deeply engrained beliefs or misinformation (Abayomi 2020). Ensuring that educational programs effectively reach and resonate with potential offenders may also prove to be difficult. Additionally, creating comprehensive legal frameworks that cover all aspects of image-based sexual abuse is complex and requires ongoing adjustments to keep up with technological advancements such as the emergence of deepfake software (Klein & Prabhakar 2024).

4 Analysis and Recommendations

While examining each strategy individually reveals certain limitations, the implementation of all five SCP techniques together could be highly effective, as they complement each other and add layers of resistance to offending. The consistent limitation across each technique is the challenge of balancing user privacy and civil rights with the surveillance and monitoring necessary for their effectiveness. Increased surveillance may drive users to adopt tools, currencies and platforms that provide anonymity, which in turn could further complicate prevention efforts (Brown 2016).

As mentioned before, the emergence of deepfake technology will require significant improvements in these measures. According to a call to action proposed by The White House (Klein & Prabhakar 2024), “Image-based sexual abuse has emerged as one of the fastest growing harmful uses of AI to date”. They propose that the government and law, as well as payment processors, cloud providers, search engines, and mobile app stores, must collaborate to develop the necessary tools and frameworks to combat its implications. For example, reducing the reward of hosting deepfake creation apps by blacklisting payment services for those sites, increasing the risk through stronger legal frameworks and consequences, and increasing the effort by restricting web and cloud services for applications marketed for creating sexual images.

Furthermore, as explained with Jaishankar’s (2008) space-transition theories’ relevance in reducing offender provocation, each of the pathways into offending must be considered when improving or implementing measures, and further research is needed to create targeted strategies for preventing IBSA and to measure their exact effectiveness. The reactive nature of many SCP strategies also calls for the need for other prevention strategies to work alongside situation crime prevention, specifically to target the root causes of IBSA, mitigate its impact on victims, and prevent re-offending.

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Harvard-UofA referencing style

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